Qatar-1b

R-0629

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_Qatar-1_230911` · created 2026-07-25T15:14:47+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460198.816 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.055
Transit depth (Rp/Rs)^2	3.32 +/- 2.01 [%]
Orbital Inclination (inc)	88.36 +/- 2.85
Airmass coefficient 1 (a1)	0.29 +/- 0.17
Airmass coefficient 2 (a2)	0.73 +/- 0.77
Scatter in the residuals of the lightcurve fit is	48.58 %
Best Comparison Star	None
Optimal Aperture	1.93
Optimal Annulus	4.75
Transit Duration (day)	0.0813 +/- 0.0077

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.182 ± 0.055 vs 0.1463 (deviation 0.6σ)
Duration fit vs published	0.0813 d vs 0.0692 d (deviation 1.45σ)
Mid-time O–C	6.7 min
Depth SNR	3.0
Residual scatter	48.58 %

Light curve

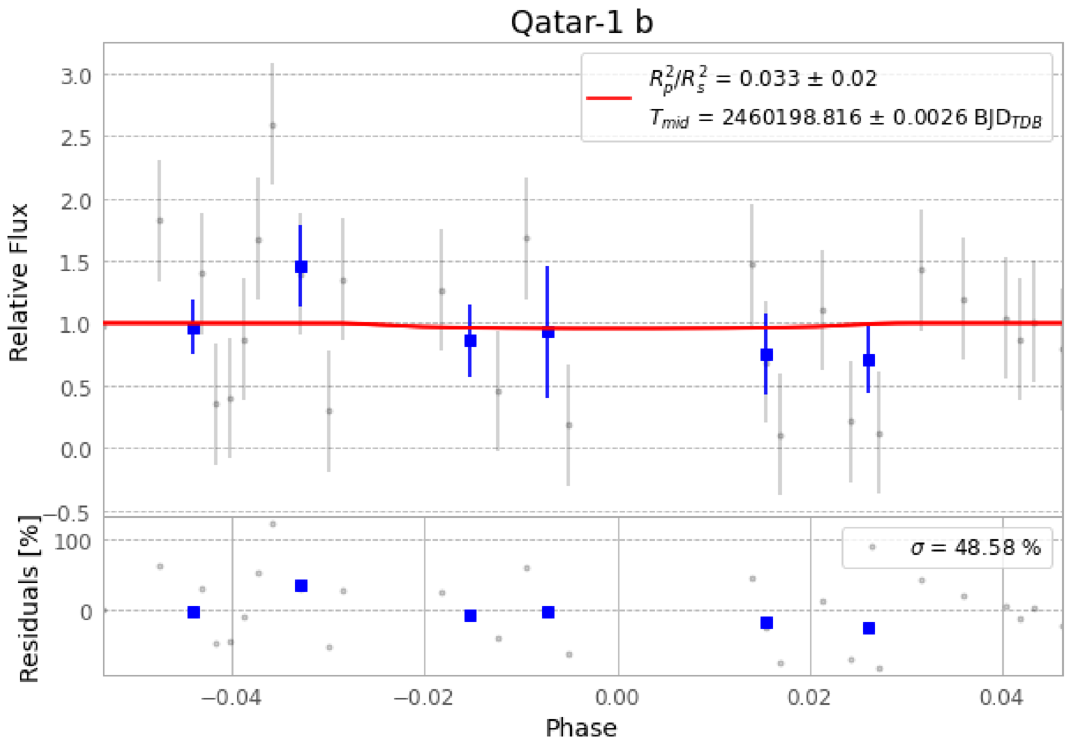

Figure 1 — FinalLightCurve_Qatar-1 b_2023-09-10.png (SHA-256 4b5a81d787d12a95…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [509, 258], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 bf1e577daf674207…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494`

Data provenance

74 FITS frames (49 MB), `Qatar-1230911053615.FITS` → `Qatar-1230911091515.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-230911.log (a89324f776a3…), FinalLightCurve_Qatar-1 b_2023-09-10.png (4b5a81d787d1…), FinalLightCurve_Qatar-1 b_2023-09-10.csv (4658052425f7…)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-25 15:14Z.